

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE – 31

WORLD CLIMATIC TYPES
THE SAVANNA OR SUDAN CLIMATE
(PART-1)

WORLD CLIMATIC TYPES

2

THE SAVANNA OR SUDAN CLIMATE

DISTRIBUTION



1

The Savanna or Sudan climate is a **Transitional type of Climate** found between the Equatorial rain forests and hot deserts.

2

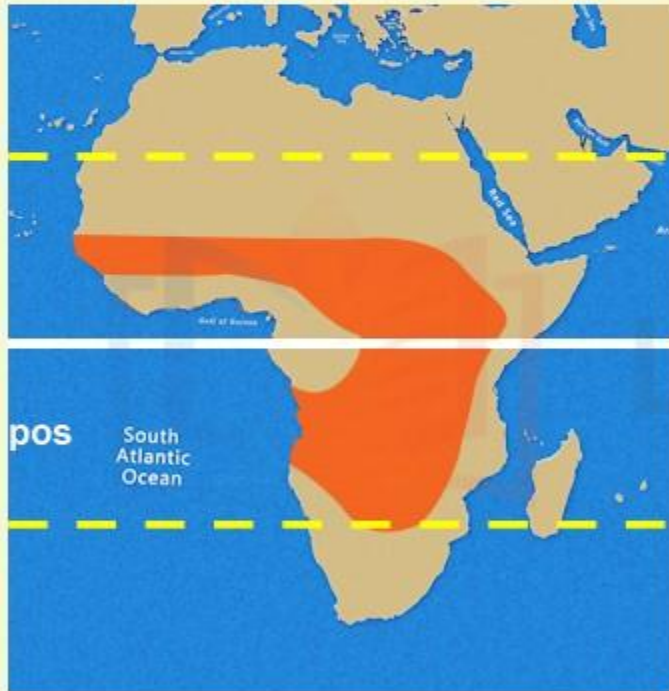
It is **confined within the tropics** and is **best developed in the Sudan Region**, where the dry and wet seasons are most distinct. Hence, it is named as the Sudan climate.

WORLD CLIMATIC TYPES

2

THE SAVANNA OR SUDAN CLIMATE

DISTRIBUTION



3

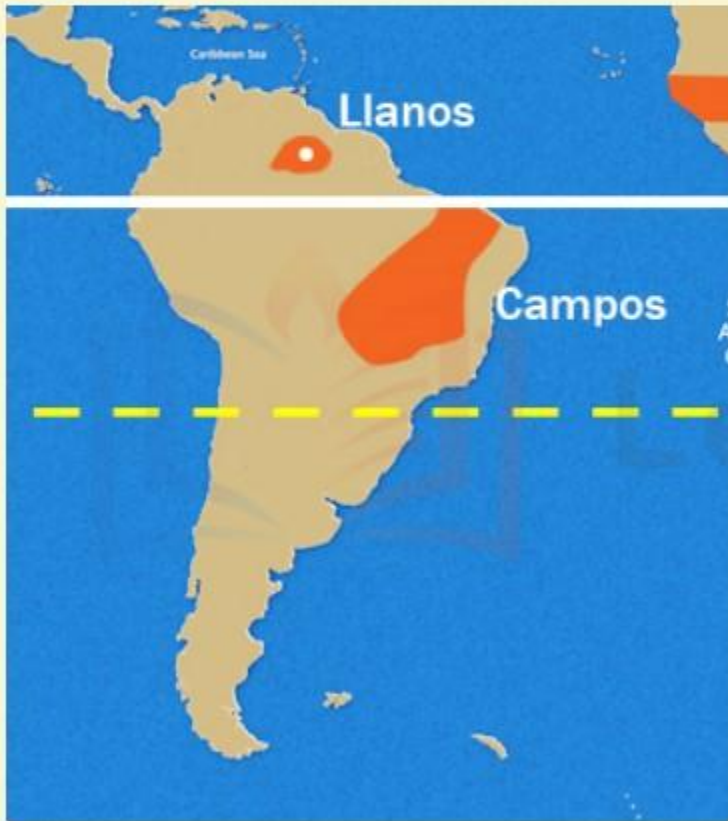
The belt includes West African Sudan and then covers southwards into East Africa and the Southern Africa, north of the Tropic of Capricorn.

WORLD CLIMATIC TYPES

2

THE SAVANNA OR SUDAN CLIMATE

DISTRIBUTION



4

In South America, there are two distinct regions of Savanna north and south of the Equator namely the Llanos of Orinoco Basin and the Campos of the Brazilian Highlands.

WORLD CLIMATIC TYPES

2

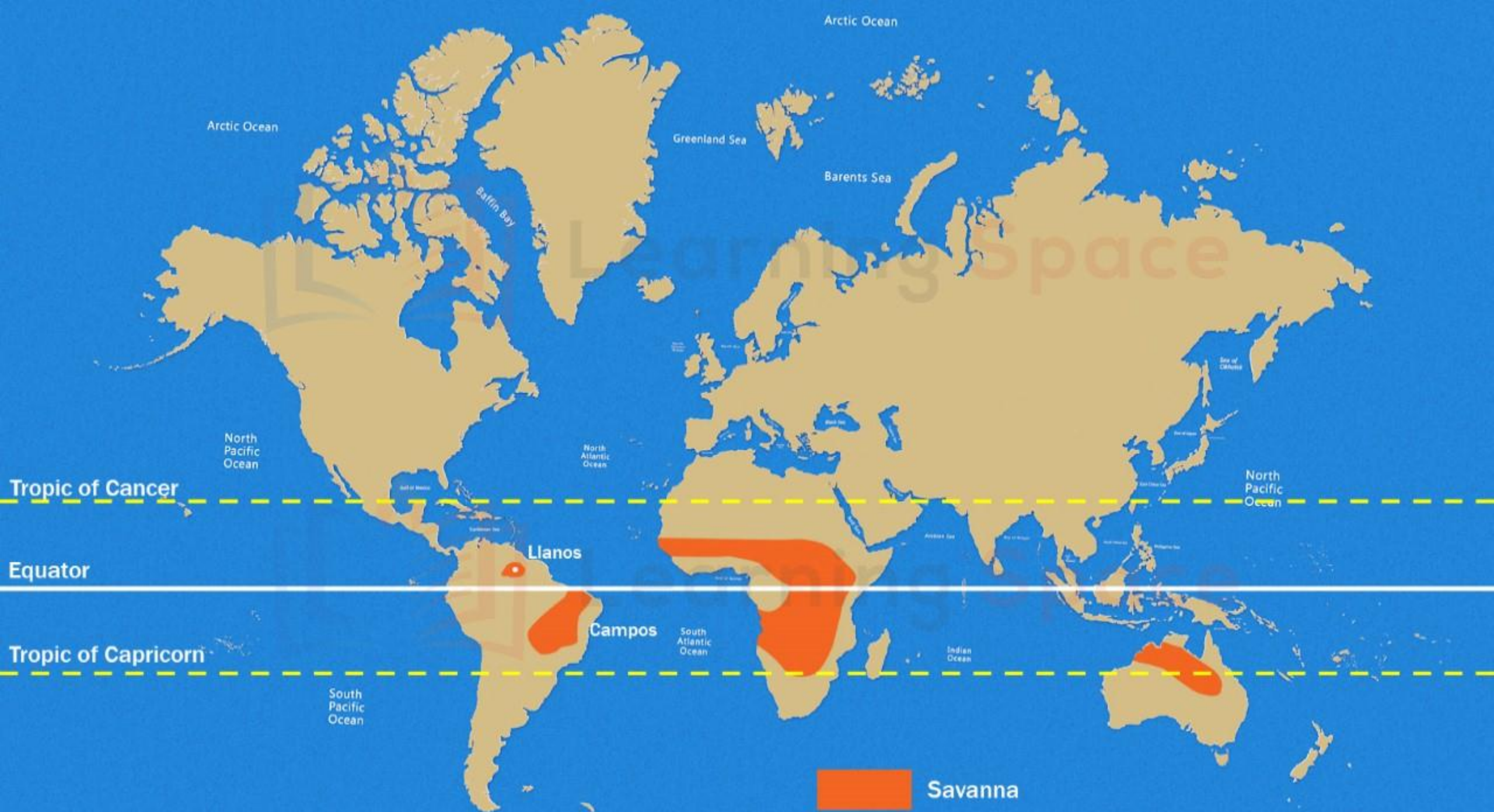
THE SAVANNA OR SUDAN CLIMATE

DISTRIBUTION



4

The Australian Savanna is located along south of the monsoon strip running from west to east north of the Tropic of Capricorn.



WORLD CLIMATIC TYPES

2

THE SAVANNA OR SUDAN CLIMATE

CLIMATE

a

RAINFALL



1

The Sudan type of Climate is characterized by an alternate hot rainy season and cool dry season.

2

The length of the rainy season decrease appreciably from the equator region pole wards towards the desert fringes. On the whole the annual precipitation is less than that of the tropical monsoon climate (around 80 to 100 cms).

WORLD CLIMATIC TYPES

2

THE SAVANNA OR SUDAN CLIMATE

CLIMATE

b

TEMPERATURE

- 1 The monthly temperature hovers between 70°F and 90°F . There is a distinct wet and dry season.

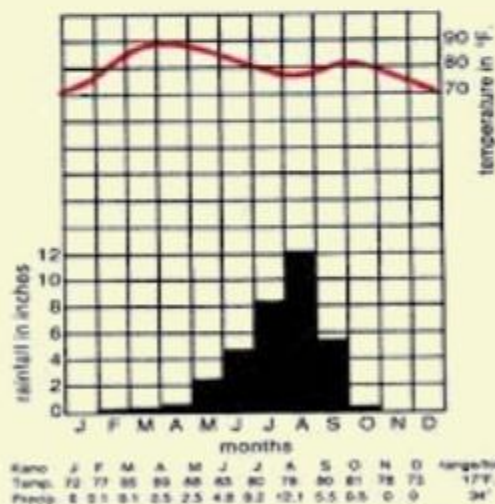


Fig. 129 (a) The Savanna or Sudan Climate in the northern hemisphere
Place: Kano, Nigeria ($11^{\circ} 58'\text{N}$, $8^{\circ} 20'\text{E}$)
Altitude: 1,539 feet
Annual precipitation: 34 inches
Annual temperature range: 17°F . (89°F – 72°F)

- 2 During the hot season, noon temperatures of over 100°F are quite frequent. When night falls the clear sky which promotes intense heating during the day causes rapid out radiation.
- 3 Temperatures drop to well below 50°F and night frost are not uncommon at this time of year.
- 4 This extreme diurnal range of temperature is another characteristic feature of the Sudan Type of Climate.

WORLD CLIMATIC TYPES

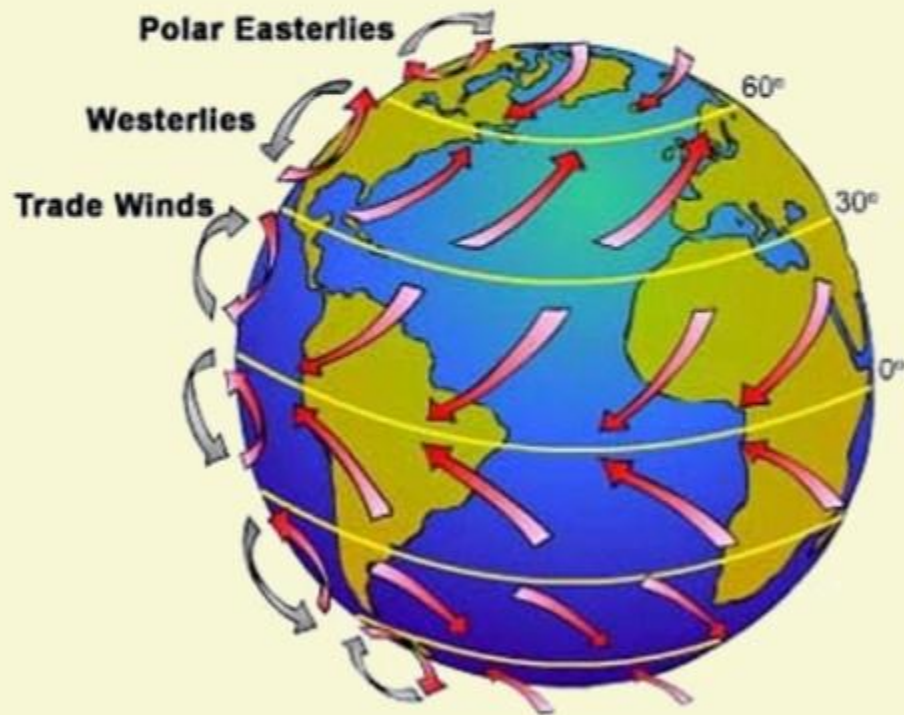
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THE SAVANNA OR SUDAN CLIMATE

CLIMATE

c

WINDS



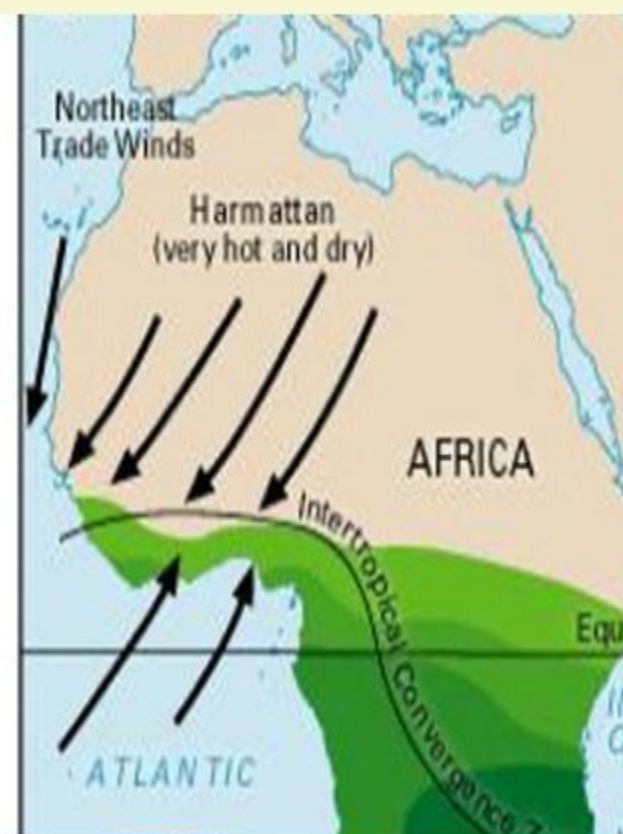
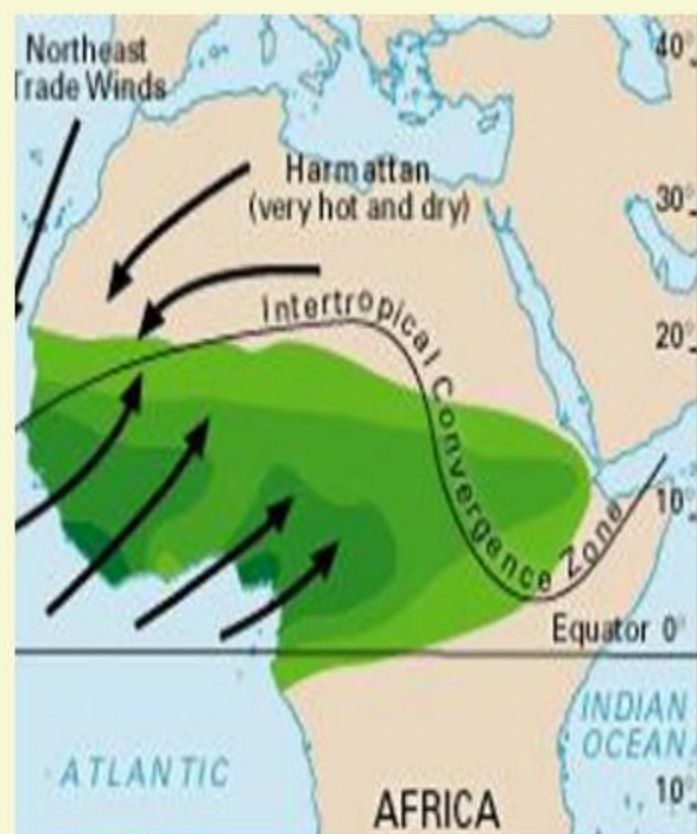
The prevailing winds of the region are the trade winds which bring rain to the coastal districts. They are strongest in the summer but relatively dry by the time they reach the continental interiors, so that grass and short trees scattered short trees predominant.



Haboob attacking Sudan

HARMATTAN

- In West Africa, the Northeast Trades in fact, blow offshore from the Sahara Desert and reach the Guinea Coast as a dry, dust laden wind called as the Harmattan meaning the Doctor. It is so dry that its relative humidity seldom exceeds 30%.
- The Harmattan provides the welcome relief from the damp air of the Guinea lands by increasing the rate of evaporation with resultant cooling effects.
- But, it is such a dry, dusty wind that besides ruining the crops it also stirs up a thick dusty haze and impedes inland river navigation.



2

THE SAVANNA OR SUDAN CLIMATE

c

WINDS

HABOOB

- The **haboob** is a strong sand-/dust storm that occurs along the **southern edges of the Sahara, the Sudan**.
- The name comes from the Arabic habb, meaning '**wind**' or '**to blow**'. The haboob is often associated with thunderstorms and even small tornadoes.
- A haboob **usually lasts about three hours and are strongest in April and May**, but occur in every month except November.
- The **city of Khartoum, Sudan experiences an average number of about 24 haboobs every year**.
- A haboob storm-front may transport and deposit large quantities of sand or dust material, which move as an extremely dense wall that can reach a height of 1000 meters (about 3,000 feet).
- The **Haboob is known by various names like the Sirocco (Mediterranean), the Khamsin (Egypt), and the Harmattan (West Africa)**.

2

THE SAVANNA OR SUDAN CLIMATE

NATURAL VEGETATION

- 1 The Savanna landscape is typified by tall grass and short trees.



- 2 It is rather misleading to call the Savanna as a tropical grassland because trees are present along with the tall grass. Therefore the terms “parkland” or “bush-veld” describes the landscape better.
- 3 Trees grow best towards Equatorial humid latitudes or along river banks but decrease in height and density away from the equator.
- 4 The trees occur as scattered individuals and they are deciduous in character this means they shed their leaves in cool dry season to prevent excessive loss of water through transpiration.

2

THE SAVANNA OR SUDAN CLIMATE

NATURAL VEGETATION

5 Others have **broad trunks with water storing devices** to survive through prolonged droughts such as Bottle trees.



BOTTLE TREES

6

Trees are mostly hard and thorny and **may exude gum like gum arabic**. Many trees are umbrella shaped exposing only a narrow edged to the strong winds.



GUM ARABIC

2

THE SAVANNA OR SUDAN CLIMATE

NATURAL VEGETATION



7

The grass grow in compact tufts and has long roots which reach down in search of water. It appears greenish and well-nourished in rainy season but turns yellow and dies down in the dry season that follows.

8

In between the tall grasses are scattered short trees and low bushes

2

THE SAVANNA OR SUDAN CLIMATE

ANIMAL LIFE



1

The Savanna particularly in Africa, is the home of wild animals. It is known as the “big game country” and thousands of animals are trapped or killed each year by people from all over the world.

2

The animals are tracked down for their skin, horns, bones or hair while others are captured alive and sent out as zoo animals, laboratory specimens or pets.

2

THE SAVANNA OR SUDAN CLIMATE

ANIMAL LIFE

3

There are two main group of animals in Savanna- **Herbivorous and Carnivorous**. The **animals are well camouflaged species** and their presence amongst the tall greenish brown grasses cannot be easily detected.



4

The Giraffe with such a long neck can located enemies a great distance away; the elephant is so huge and strong that few animals will come closer to it.



2

THE SAVANNA OR SUDAN CLIMATE

ANIMAL LIFE

5

Animals like tiger and leopard have stripes or dots on them which perfectly match the background of the savanna. They often hide themselves in shady spots amidst the tall bushes and many animals are caught unawares by these carnivorous animals.



2

THE SAVANNA OR SUDAN CLIMATE

ANIMAL LIFE

6

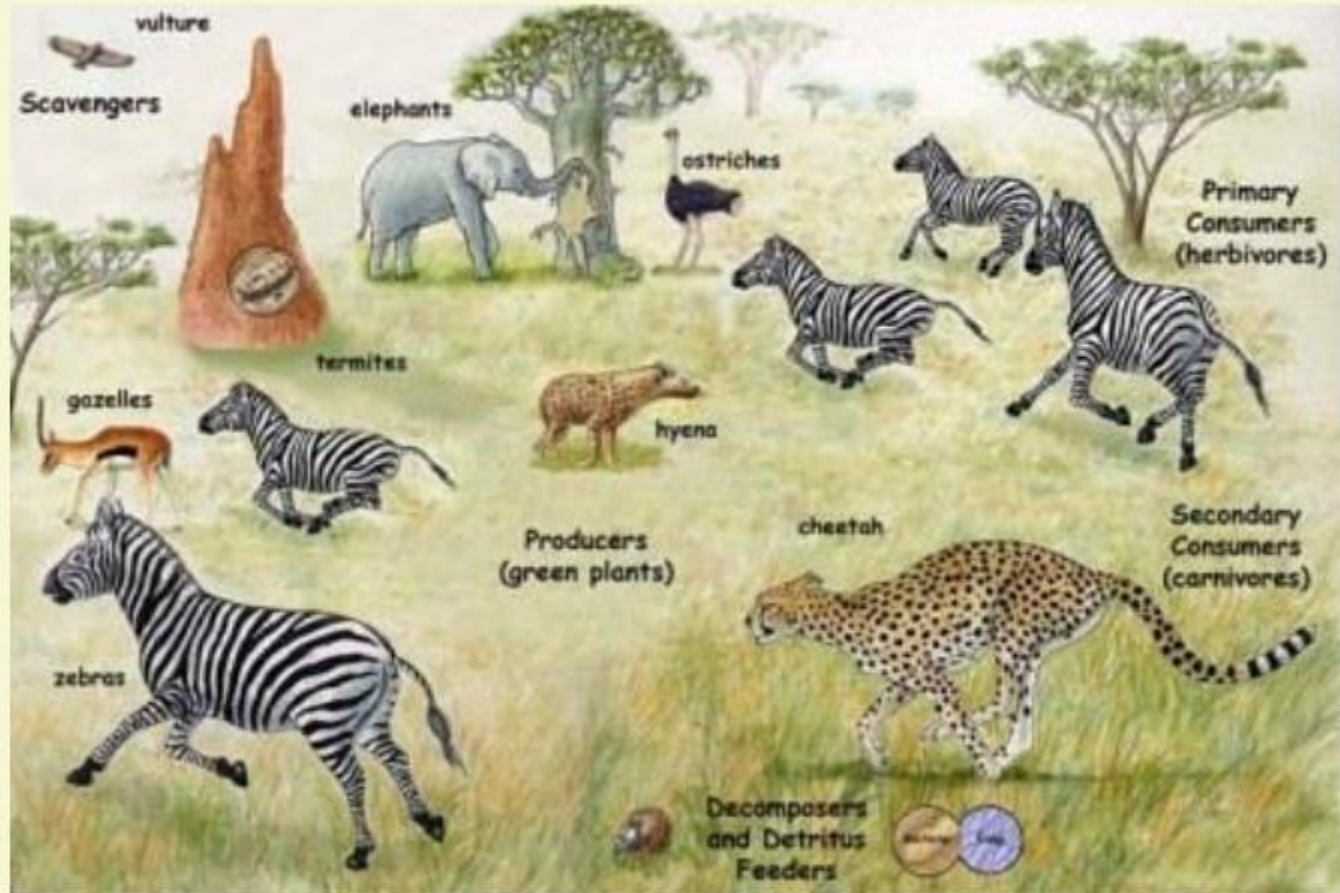
Along rivers and marshy lakes, numerous species of reptiles and mammals are found which includes crocodile, alligators, giant lizards, rhinoceros and hippopotamus.



2

THE SAVANNA OR SUDAN CLIMATE

ANIMAL LIFE



7

Because of this huge diverse species of animals Savanna region can be said to be an “animal paradise”.

8

In many parts of Africa National Parks have been setup to control the killing of animals and this invites large number of tourist to see the animals.

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For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

OUR TEAM

G. V. Rao

K. Srikanth

D. Sunil Kumar

L. V. Krishna

D. Chaitanya

Amrita Naidu

M. Binukrishna

G. Ravi Babu

D. Joji

K. Victor Babu

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